Analysis of a Tensile Coupon with a Notch

Figure 1 Coupon

The coupon is 1 unit long, 0.2 units high. The notch starts at 0.4 ends at 0.6, and is 0.1 high.
Two dimension model. X along the length, y is up. Left end fixed, right end has enforced motion in the x direction.

Quad4 mesh, with “p:2” elements in Elmer, Mumps direct solver

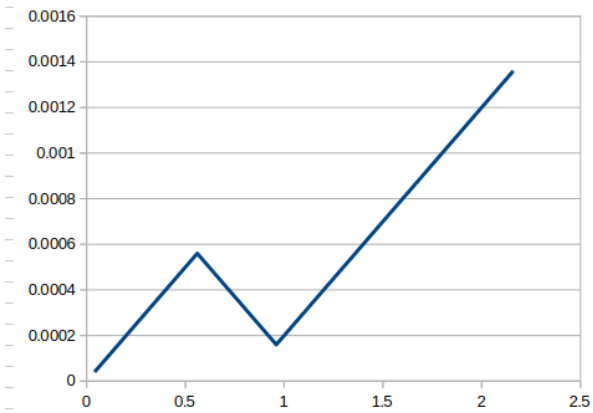
Loading is variable with time the right side is pulled as a function of the time variables. Instead of automatic time stepping control, direct cut back time is used by having multiple time step sizes, the type is scanning because transient does not support the UMAT material model.

Simulation Type = Scanning
timestep intervals(4) = 4 20 40 60
timestep sizes(4) = 0.04 0.02 0.01 .02
output intervals(4) = 1 1 1 1

Displacement 1 = Variable Time; Real MATC "0.001*tx"

Cause a displacement history of the table on the left, a second run is done with an unload/reload curve on the right

Time	Displacement
0.00000	0.00000
0.16000	0.00160
0.56000	0.00190
0.96000	0.00160
2.16000	0.00216

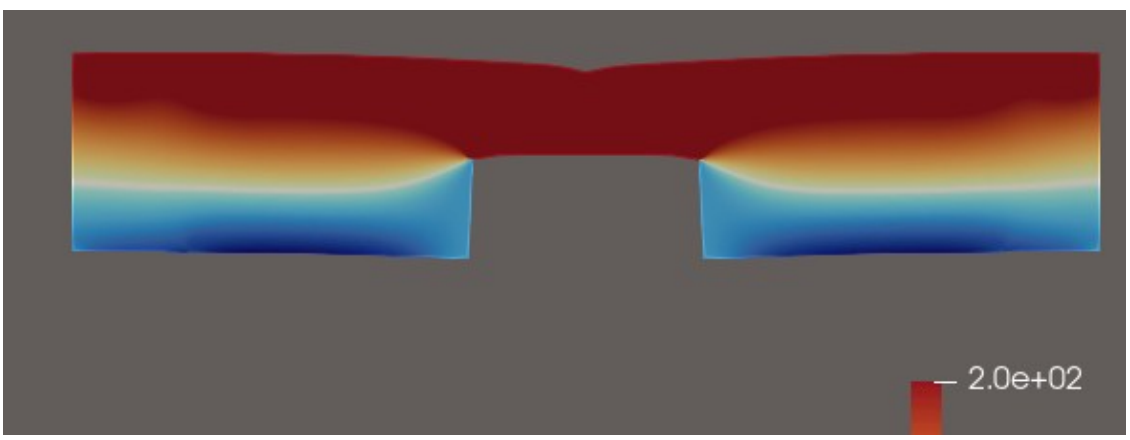


Material Model

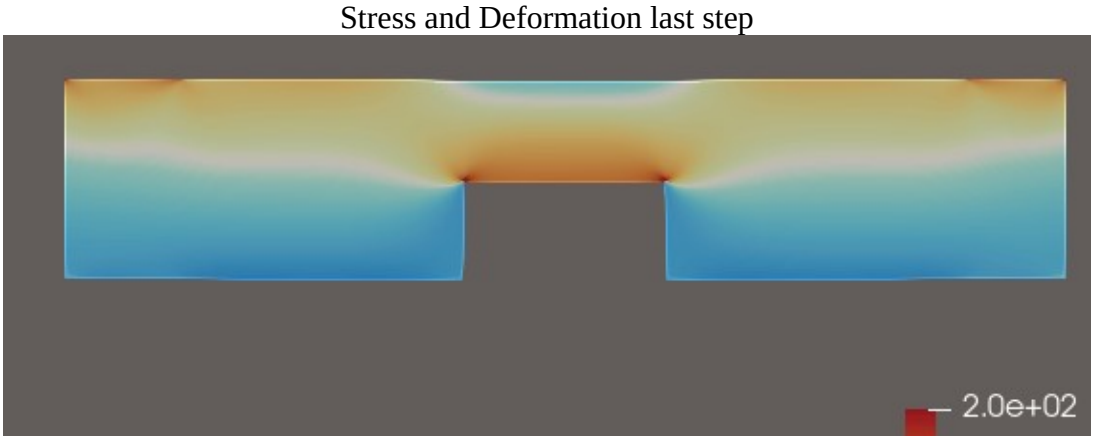
UMAT is used to incorporate the mfront plasticity model using stress-strain point to define the stress-strain curve

```
Material 1
! UMAT test
  Number of Material Constants = Integer 1
  Number of State Variables = Integer 5
  ! List material constants as {}:
 Material Constants(1) = Real 270.0
  Density = 0.0
  Reference Temperature = 270.0
  UMAT Subroutine = File "/home/xxxx/tfel/src/libABAQUSBEHAVIOUR"
"USERDEFINEDVISCOPLASTICITY_PSTRAIN"
  Name = "USERDEFINEDVISCOPLASTICITY_PSTRAIN"  ! This specifies the CMNAME
argument of UMAT!
End
```

Results Loading case



Stress and Deformation last step



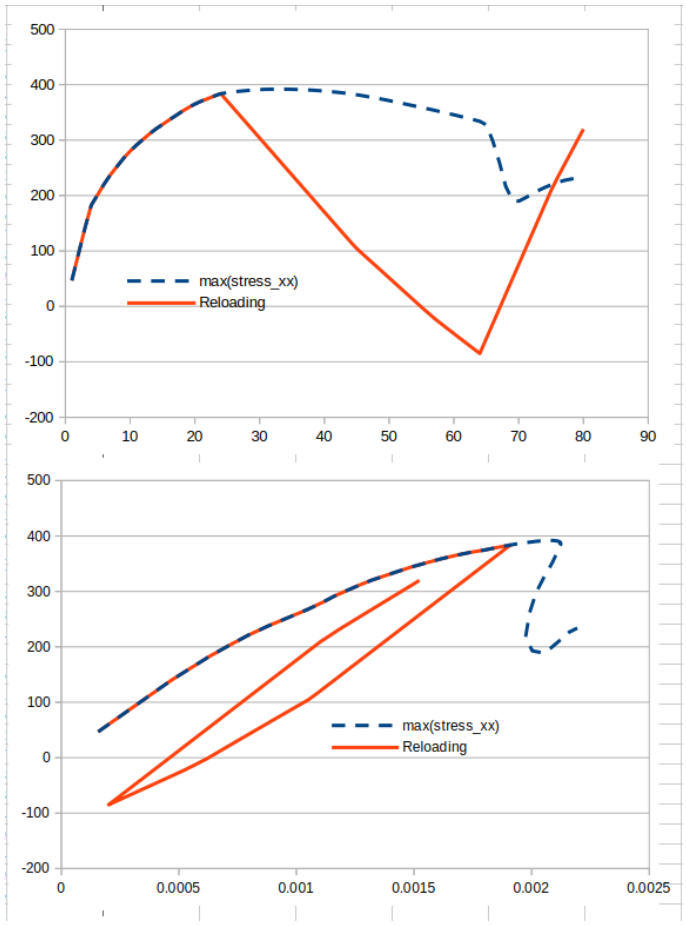
Stress and Deformation Step 10

Animation

Unloading and reloading case
Same as above until the re-loading starts.

Animation

Stress Strain Results at top left corner of the notch, the top curve is stress at the intervals, the second curve is the output stress strain curve.



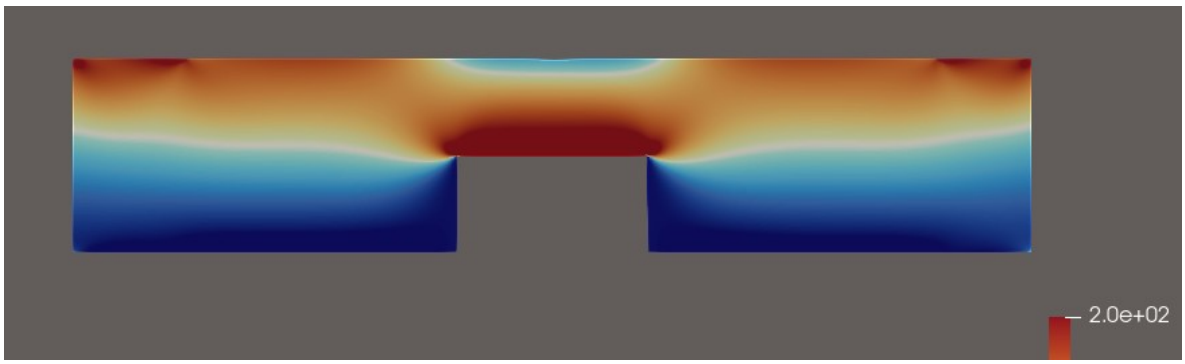
Future testing:

Automatic time step control. The available one in Elmer doesn't appear to work with this model. The direct solution and enforced displacement results in a solution norm of 0.0000 at the end of every step.

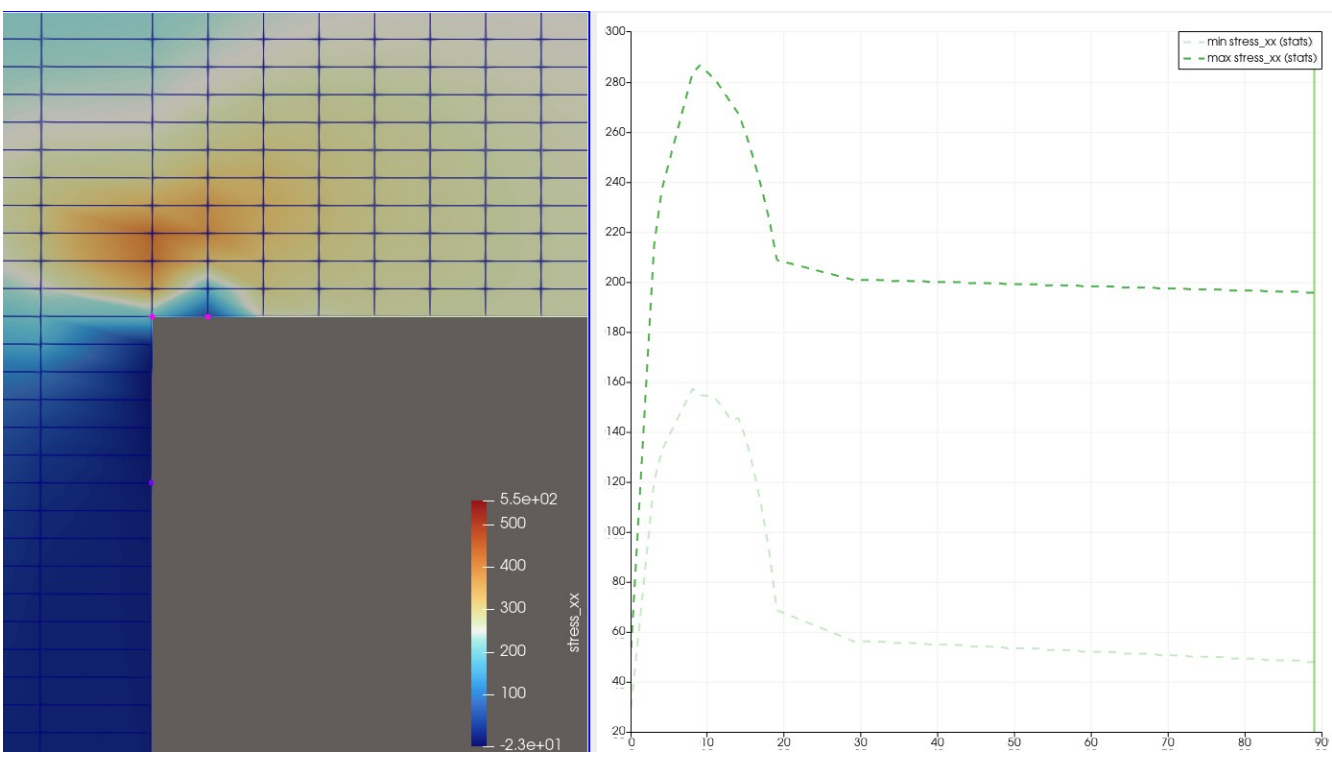
The default criterion is that the norms of the solutions to each system of field equations do not differ more than the given threshold value. If the solution norm is zero, than for this problem the default will have no effect.

Damage Model:

Initial test with a damage model requires more cut back in time steps than the plasticity model

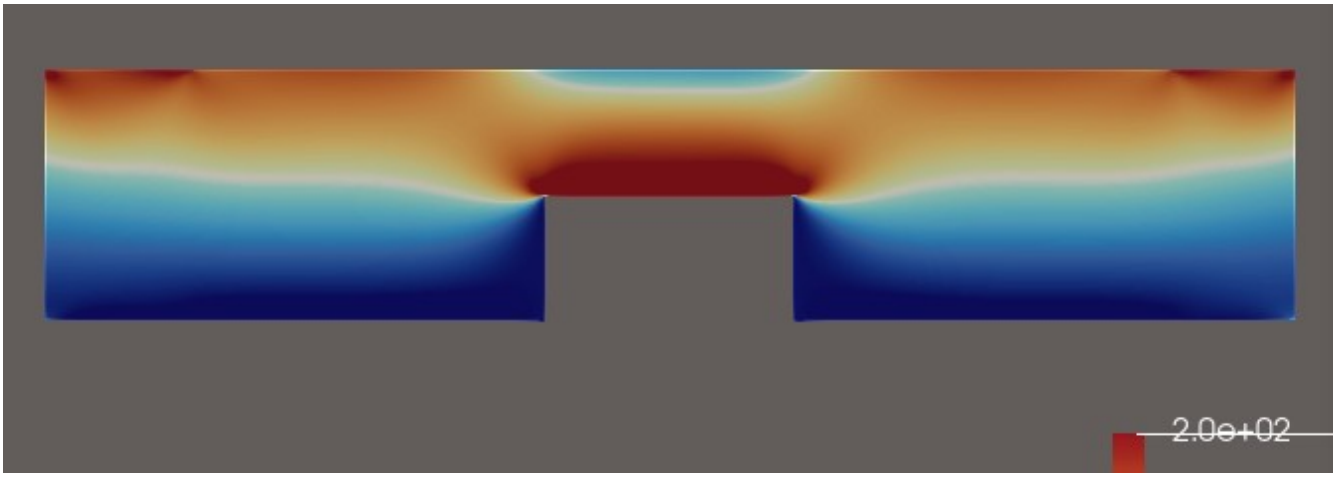


Stress at last time step

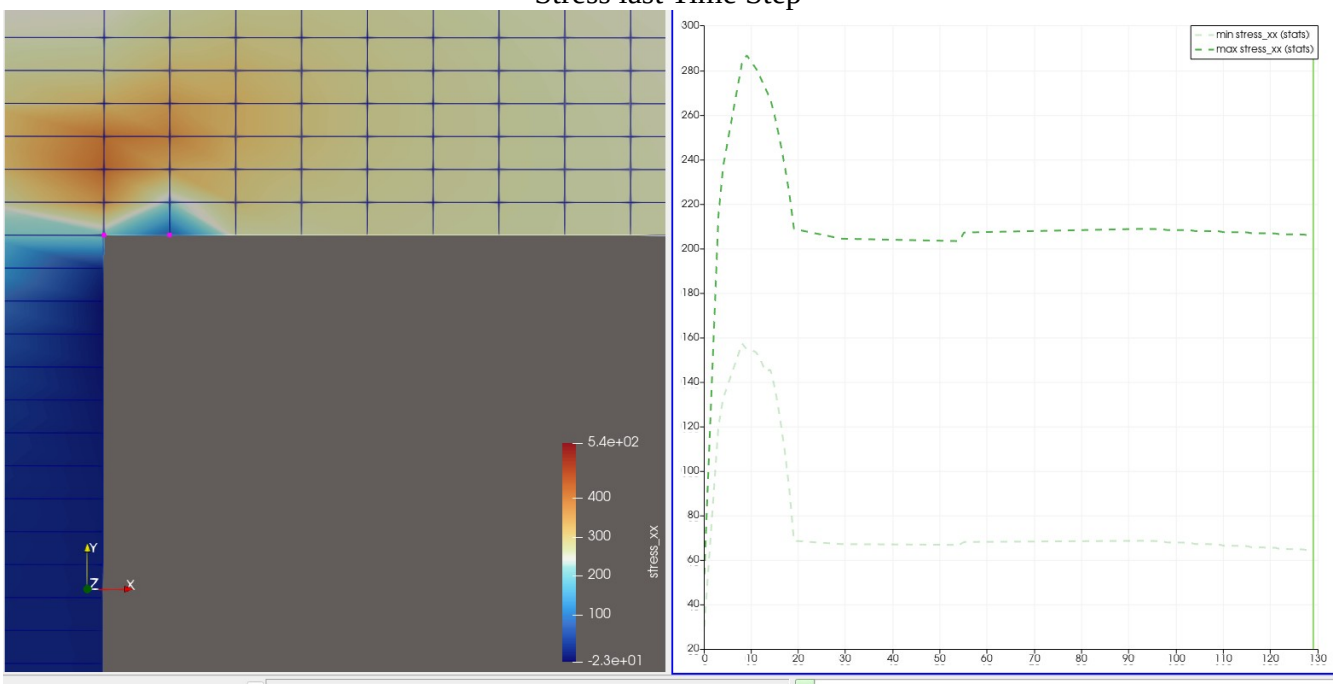


Stress Time history at notch and just forward of notch

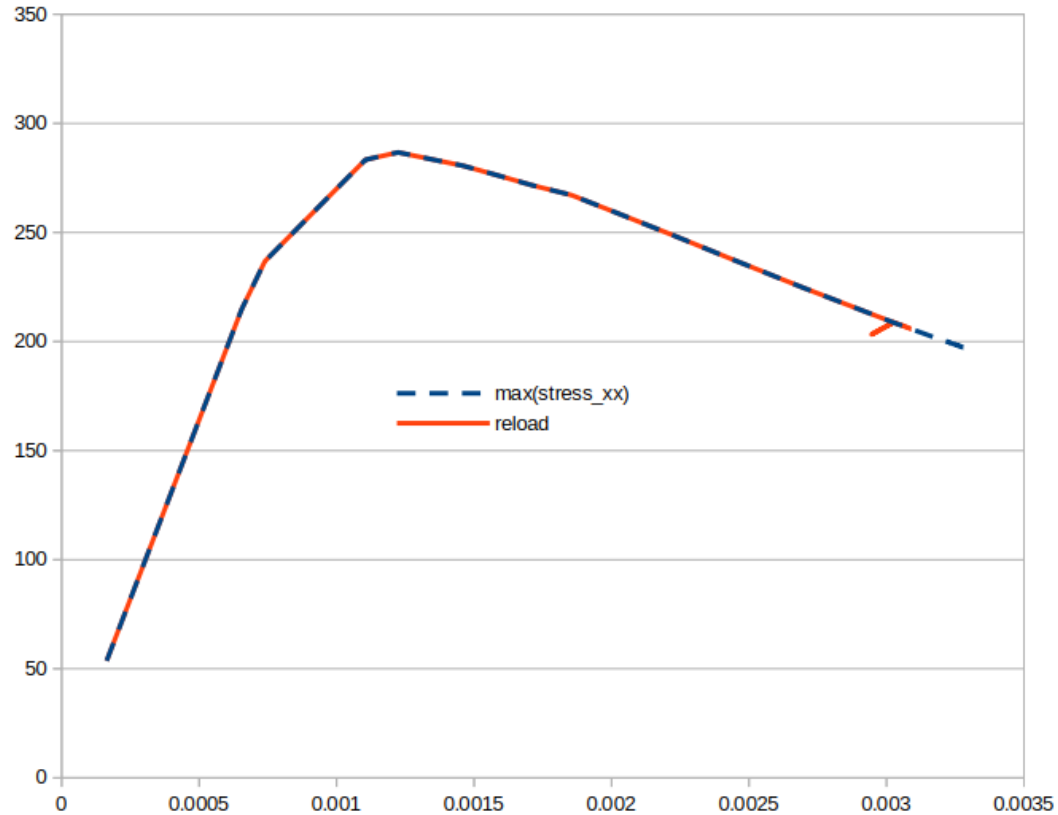
The damage model with unloading and reloading



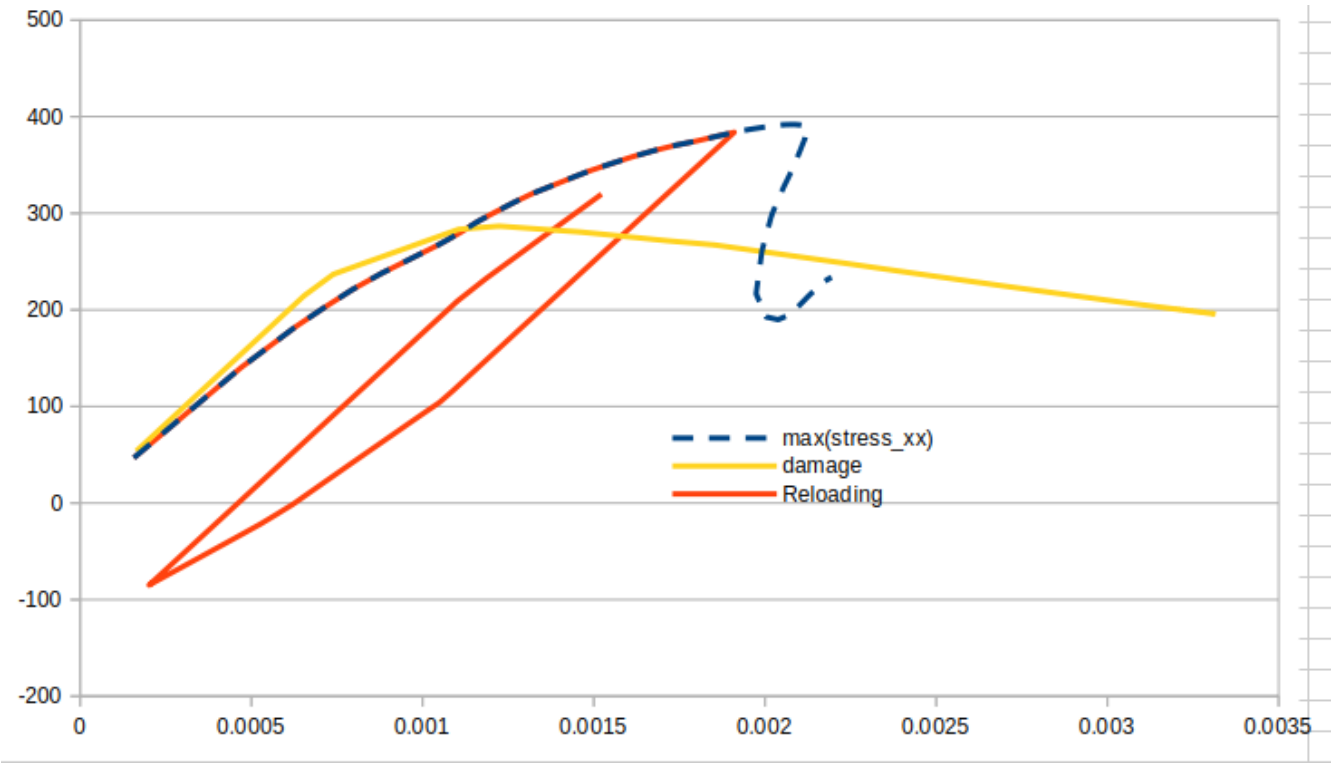
Stress last Time Step



Time History



Stress Strain Curve



Stress Strain Comparison to Damage Model

Cracking/Removal of Damaged/Broken Elements:

In ABAQUS removal of elements above a certain strain is an automated option. In Calculix it is possible but not as automated.

Continuation to cracking/breaking could be simulate if elements could be removed based on strain value.

Elmer has active/passive elements but the examples and test problems are based on inputs not outputs. The active/passive would have to be based on each elements strain. Then cracking or breaking could start to be simulated.